Excel Pivot Tables

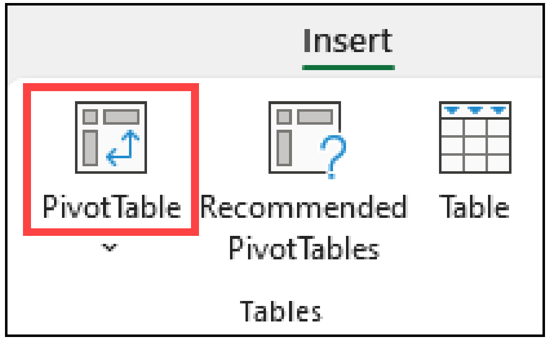
Objectives

- Summarize data using Microsoft Excel Pivot Table
- Analyze data using Microsoft Excel Pivot Table
- Explore data using Microsoft Excel Pivot Table
- Present data using Microsoft Excel Pivot Table

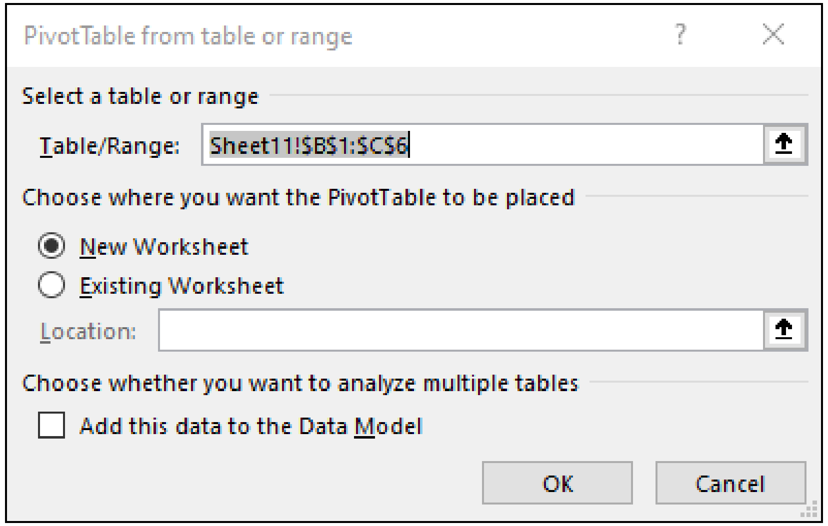
Pivot Tables

- A **Pivot Table** is a powerful tool in Excel used to calculate, summarize, and analyze data. It allows you to see comparisons, patterns, and trends in your data. Pivot Tables function slightly differently depending on the platform you are using to run Excel.

Create a PivotTable in Excel for Windows



1. Select the cells you want to create a Pivot Table from.
2. Select **Insert** > **PivotTable**.
3. This creates a PivotTable based on an existing table or range.
4. Choose where you want the PivotTable report to be placed. Select **New Worksheet** to place the PivotTable in a new worksheet or **Existing Worksheet** and select where you want the new PivotTable to appear.
5. Select OK.



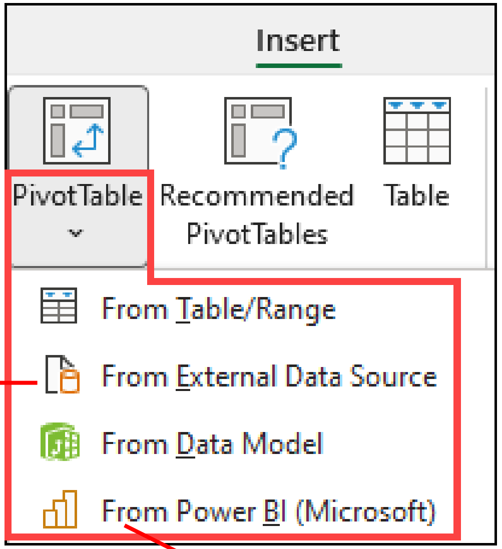
Data format tips and tricks

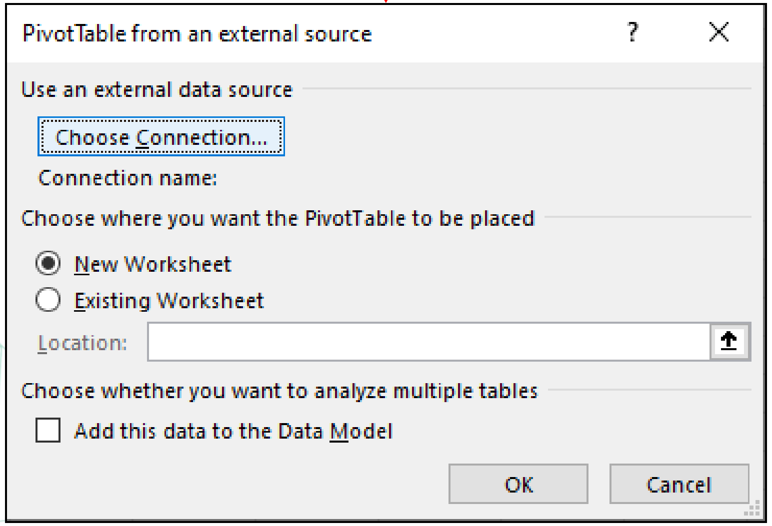
	A	B	C	D	E	F
1	Year	Category	Product	Sales	Rating	
2	2017	Components	Chains	$20,000	75%	
3	2015	Clothing	Socks	$ 3,700	22%	
4	2017	Clothing	Bib-Shorts	$ 4,000	22%	
5	2015	Clothing	Shorts	$13,300	56%	
6	2017	Clothing	Tights	$36,000	100%	
7	2015	Components	Handlebars	$ 2,300	35%	
8	2016	Clothing	Socks	$ 2,300	28%	
9	2016	Components	Brakes	$ 3,400	36%	
10	2016	Bikes	Mountain Bikes	$ 6,300	40%	
11	2017	Components	Brakes	$ 5,400	38%	
12	2016	Accessories	Helmets	$17,000	90%	
13	2016	Accessories	Lights	$21,600	90%	
14	2016	Accessories	Locks	$29,800	90%	

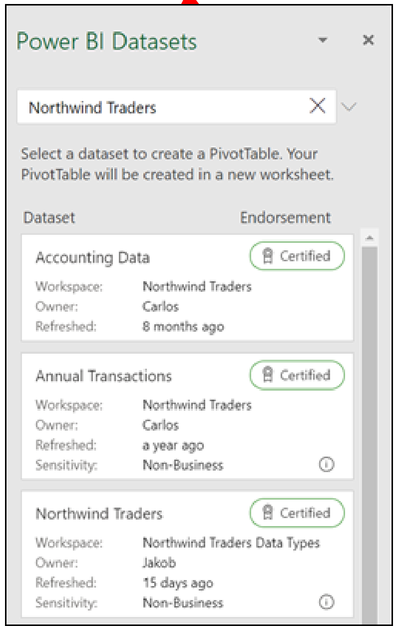
•**Use clean, tabular data** for best results.
•Organize your data in **columns**, not rows.
•**Ensure all columns have headers** with a single row of unique, non-blank labels for each column. Avoid double rows of headers or merged cells.
•**Format your data as an Excel table**: Select anywhere in your data, and then select **Insert > Table** from the ribbon.
•If you have complicated or nested data, **use Power Query** to transform it (for example, to **unpivot your data**) so it's organized in columns with a single header row.

Create a PivotTable From Other Sources

1. By clicking the down arrow on the button, you can select from other possible sources for your PivotTable. In addition to using an existing table or range, there are three other sources you can select from to populate your PivotTable.

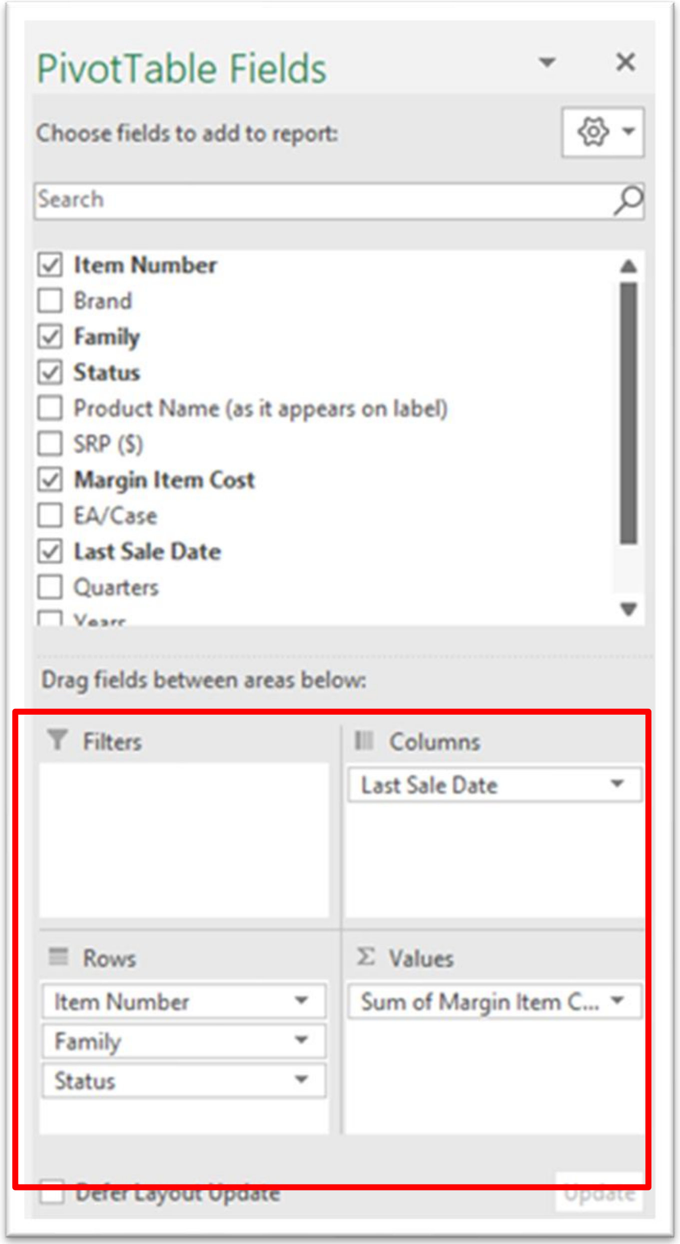






Build out your PivotTable

- To add a field to your PivotTable, select the field name checkbox in the **PivotTables Fields** pane.
- To move a field from one area to another, drag the field to the target area.

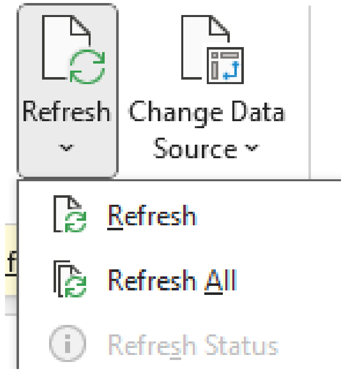


Note: Selected fields are added to their default areas: non-numeric fields are added to **Rows**, date and time hierarchies are added to **Columns**, and numeric fields are added to **Values**.

Refreshing PivotTables

•**Single PivotTable**: If you add new data to your PivotTable data source and need to refresh just one PivotTable, right-click anywhere in the **PivotTable range**, and then select **Refresh**.
•**Multiple PivotTables**: If you have multiple PivotTables and want to refresh all of them:

- Select any cell in any PivotTable.
- On the ribbon, go to **PivotTable Analyze**.
- Click the arrow under the **Refresh** button.
- Select **Refresh All**.



Refreshing PivotTables

Warning: Data Format Consistency

•**Potential Issues**: If the data formats change (e.g., from text to number, date format changes, etc.), it may lead to errors during the refresh.

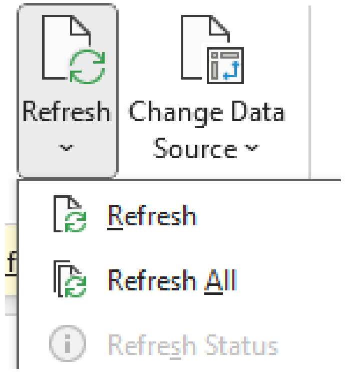
•**Best Practices**:

- Ensure that the data formats remain consistent in the source data.
- Before refreshing, check for any changes in the data format to avoid unexpected results or errors.

Action Steps:

•Regularly validate the data types in your source data.

•Keep documentation of any changes made to the data structure.



Working with PivotTables Values

Summarize Values By

•By default, PivotTable fields placed in the **Values** area are displayed as a **SUM**. If Excel interprets your data as text, it will be displayed as a **COUNT**. Therefore, it is crucial to ensure that you don't mix data types for value fields.

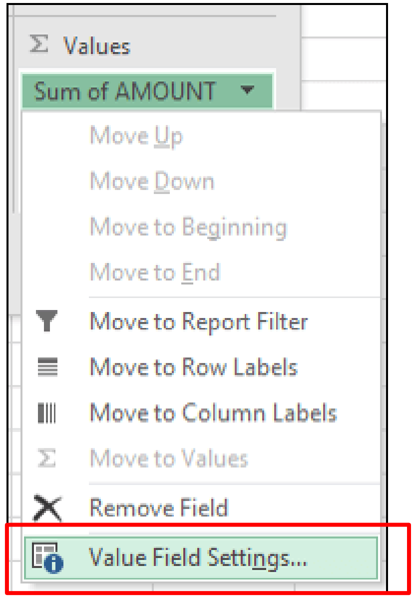
•To change the default calculation:

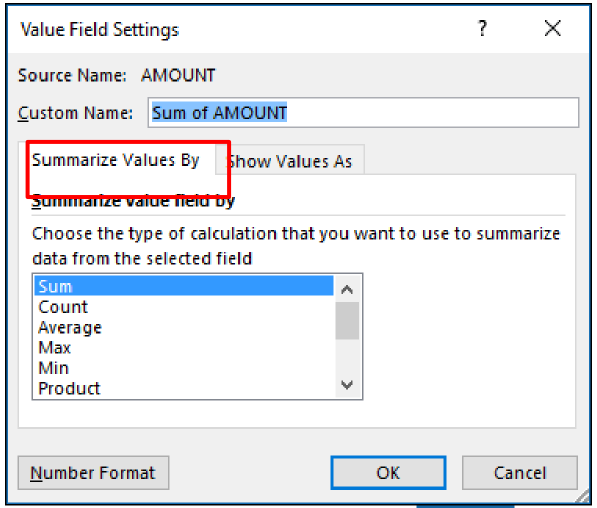
- Select the **arrow** to the right of the field name in the Values area.
- Choose **Value Field Settings** from the dropdown menu.
- In the **Summarize Values By** section, select your preferred calculation method (e.g., **Average**, **Max**, **Min**, etc.).

•Note that when you change the calculation method, Excel automatically appends it in the **Custom Name** section, such as "Sum of FieldName." You can modify this name if needed.

•To change the number format for the entire field:

- Select **Number Format** within the **Value Field Settings** dialog box.
- Choose your desired number format and apply it.





Working with PivotTables Values

Show Values As

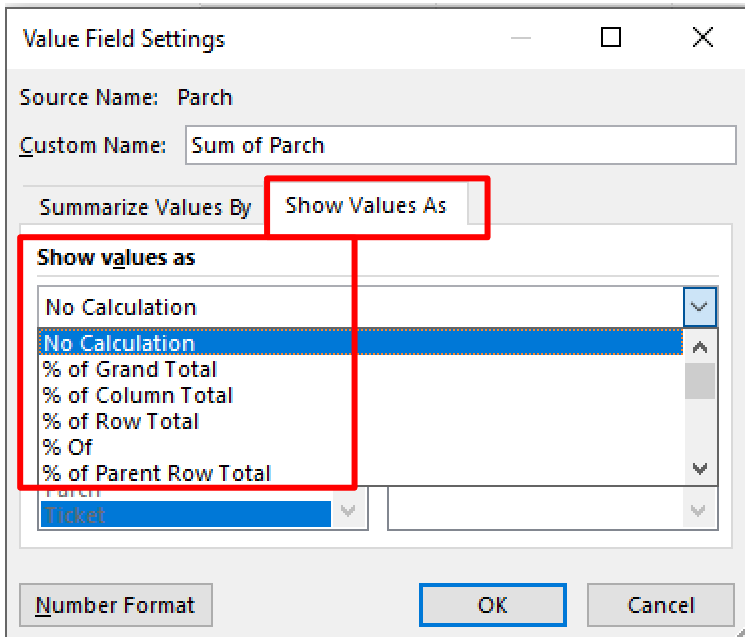
•**Display as Percentage**: Instead of summarizing data with a calculation, you can display it as a **percentage** of a field. For example, you might change household expense amounts to display as a **% of Grand Total** instead of the sum of the values.

•To configure this:

- Open the **Value Field Settings** dialog box.
- Select the **Show Values As** tab.
- Choose the **percentage** option you want (e.g., **% of Grand Total**, **% of Column Total**, etc.).

•**Display Both Calculation and Percentage**:

- To show a value as both a calculation and a percentage, drag the item into the **Values** section twice.
- Set the **Summarize Values By** option for the first instance and configure the **Show Values As** option for the second instance to display it as a percentage.



AMOUNT	MONTH			
CATEGORY	January	February	March	Grand Total
Entertainment	5.10%	6.38%	6.13%	17.61%
Grocery	12.00%	12.25%	13.27%	37.52%
Household	8.93%	11.49%	10.21%	30.63%
Transportation	3.78%	5.87%	4.59%	14.24%
Grand Total	29.81%	35.99%	34.20%	100.00%

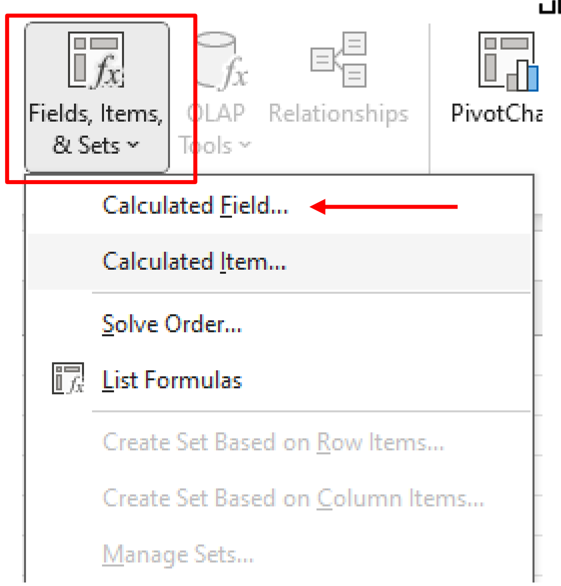
Create formulas in a PivotTable

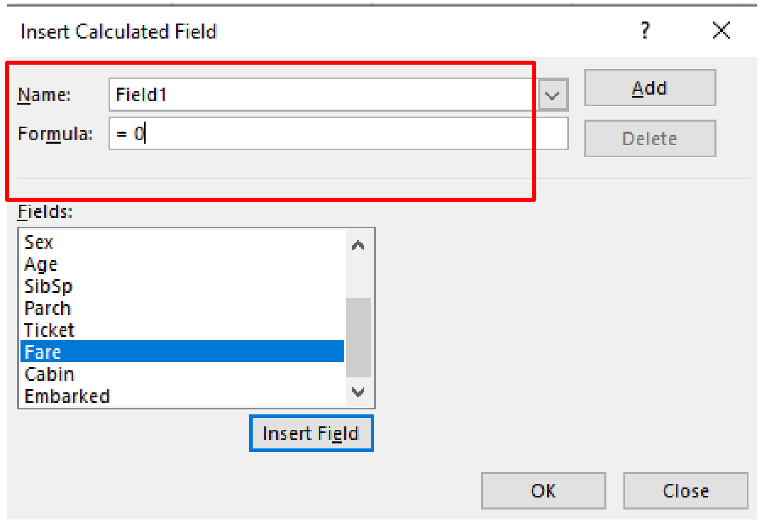
- **Calculated Field vs. Calculated Item**:
  - **Calculated Field**: Use this when you want to create a formula that utilizes data from other fields in the PivotTable.
  - **Calculated Item**: Use this when you need a formula that works with data from one or more specific items within a field.
- **Calculated Items**:
  - You can enter different formulas for each cell.
  - For example, if you have a calculated item named **OrangeCounty** with a formula of =Oranges * 0.25 across all months, you can modify the formula to =Oranges * 0.5 specifically for June, July, and August.
- **Adjusting Order of Calculation**:
  - If you have multiple calculated items or formulas, you can adjust the order in which they are calculated to ensure accurate results.

Create formulas in a PivotTable

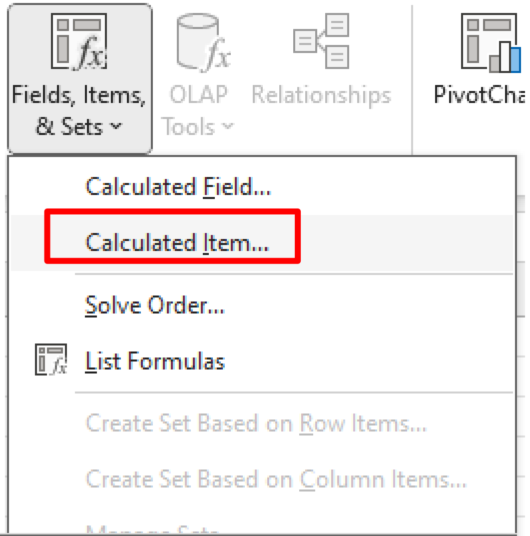
Add a calculated field

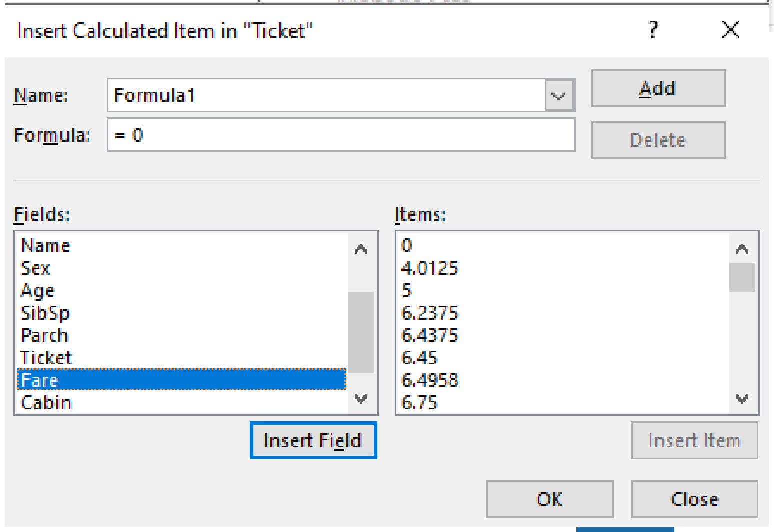
1. Click the PivotTable.
2. This displays the PivotTable Tools, adding the **Analyze** and **Design** tabs.
3. On the **Analyze** tab, in the **Calculations** group, click **Fields, Items, & Sets**, and then click **Calculated Field**.
4. In the **Name** box, type a name for the field.
5. In the **Formula** box, enter the formula for the field.
6. To use the data from another field in the formula, click the field in the **Fields** box, and then click **Insert Field**. For example, to calculate a 15% commission on each value in the Sales field, you could enter **= Sales * 15%**.
7. Click **Add**





Create formulas in a PivotTable



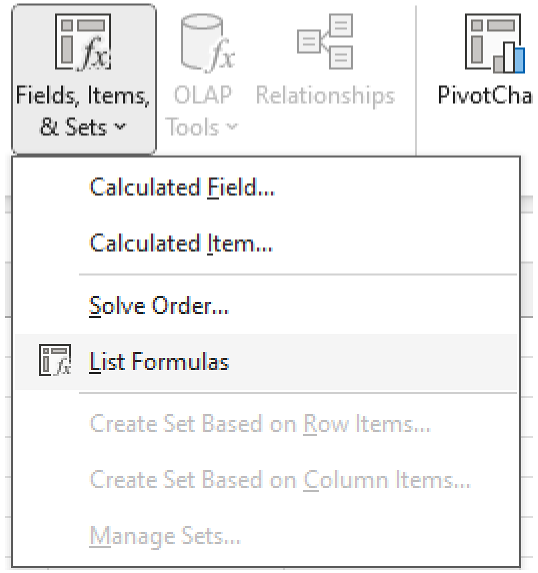


Add a calculated item to a field

1. Click the PivotTable.
2. This displays the PivotTable Tools, adding the **Analyze** and **Design** tabs.
3. If items in the field are grouped, on the **Analyze** tab, in the **Group** group, click **Ungroup**.
4. Click the field where you want to add the calculated item.
5. On the **Analyze** tab, in the **Calculations** group, click **Fields, Items, & Sets**, and then click **Calculated Item**.
6. In the **Name** box, type a name for the calculated item.
7. In the **Formula** box, enter the formula for the item.
8. To use the data from an item in the formula, click the item in the **Items** list, and then click **Insert Item** (the item must be from the same field as the calculated item).
9. Click **Add**.

View all formulas that are used in a PivotTable

1. Click the PivotTable.
2. This displays the PivotTable Tools, adding the **Analyze** and **Design** tabs.
3. On the **Analyze** tab, in the **Calculations** group, click **Fields, Items, & Sets**, and then click **List Formulas**.



Edit a PivotTable formula

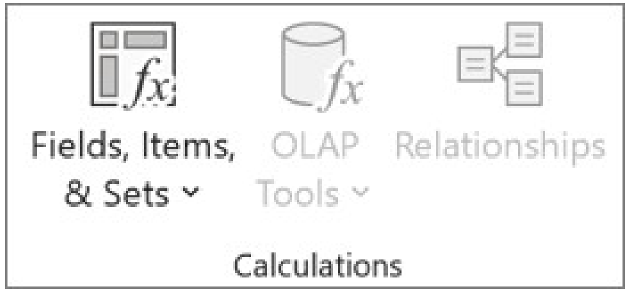
1.Determine the Formula Type:
- Identify whether the formula is part of a **calculated field** or a **calculated item**.

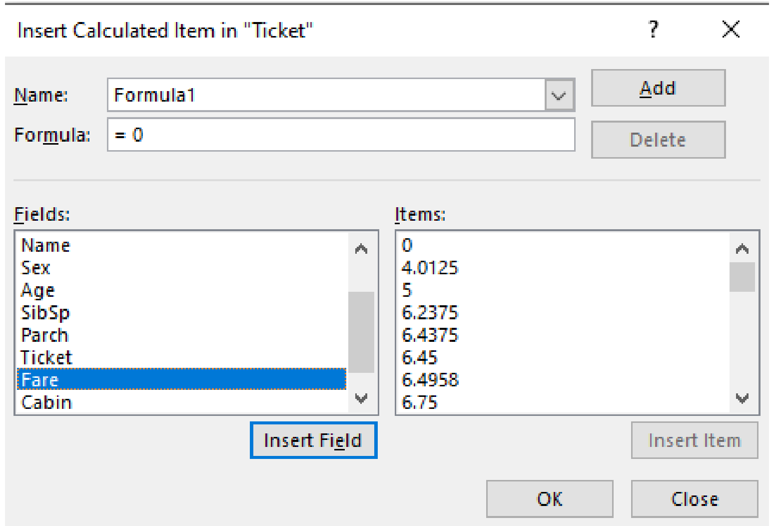
2.Click the PivotTable:
- This action displays the **PivotTable Tools**, adding the **Analyze** and **Design** tabs to the ribbon.

3.Open List Formulas:
- On the **Analyze** tab, in the **Calculations** group, click **Fields, Items, & Sets**.
- Then select **List Formulas** from the dropdown menu.

4.Locate the Formula:
- In the list of formulas, find the one you want to change under **Calculated Field** or **Calculated Item**.
- For **calculated items**, if there are multiple formulas, the default formula will show the calculated item name in column **B**. Additional formulas will display both the calculated item name and the names of intersecting items.
- **Example**: You might have a default formula for a calculated item named **MyItem**, and another formula for this item identified as **MyItem January Sales**. In the PivotTable, this formula would appear in the **Sales** cell for the **MyItem** row and **January** column.

5.Edit the Formula:
- Use one of the following methods to modify the formula (not specified in the provided text; generally, involves selecting the formula and making changes directly or using the formula editor).



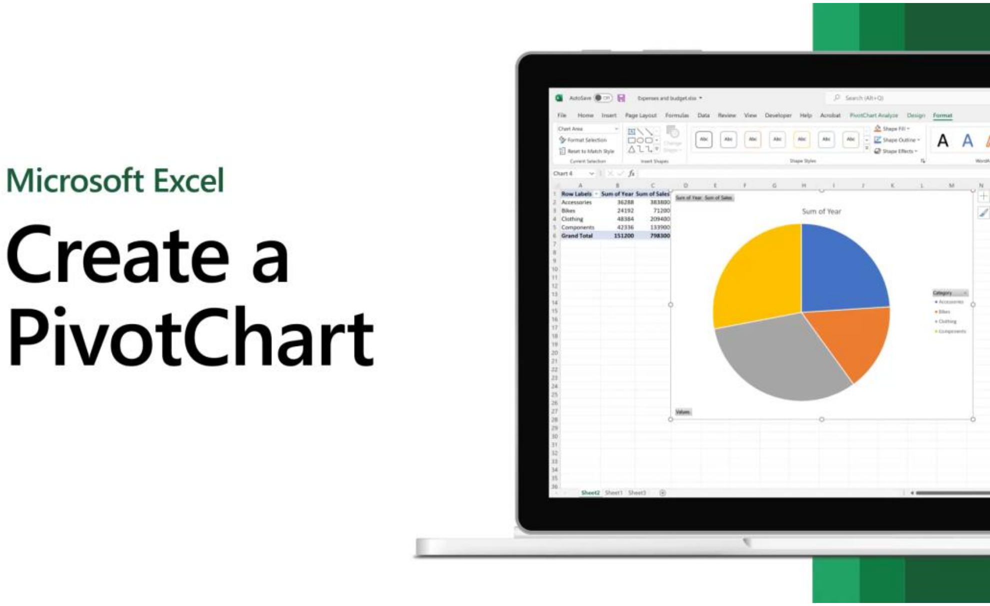


Create a PivotChart

Create a PivotChart

Sometimes it's hard to see the big picture when your raw data hasn’t been summarized. Your first instinct may be to create a PivotTable, but not everyone can look at numbers in a table and quickly see what's going on. Pivot Charts are a great way to add data visualizations to your data.

1. Select a cell in your table.
2. Select **Insert** and choose Pivot Chart button **PivotChart**.
3. Select where you want the PivotChart to appear.
4. Select OK.
5. Select the fields to display in the menu.



Create a chart from a PivotTable

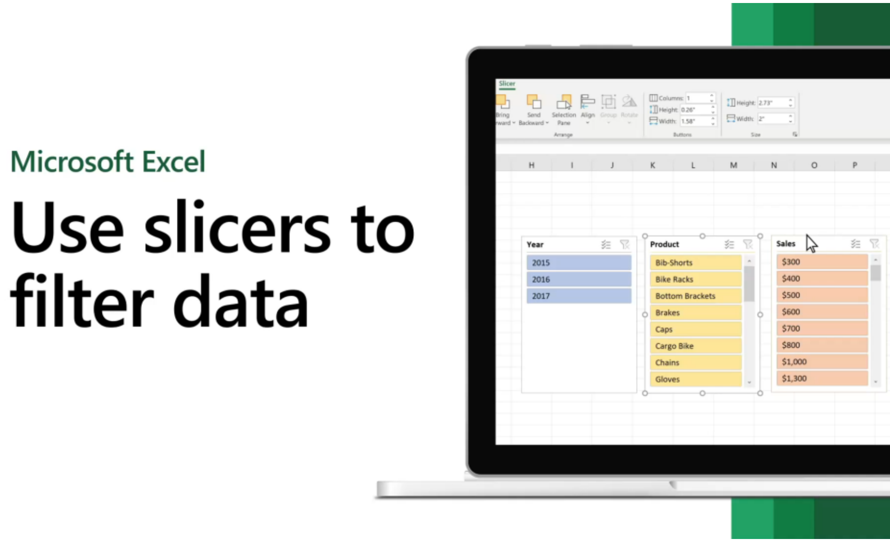
1. Select a cell in your table.
2. Select **Insert** and choose **PivotChart**.
3. Select a chart.
4. Select **OK**.

Create a Slicer

Use slicers to filter data

You can use a slicer to filter data in a table or PivotTable with ease.

1. Click anywhere in the table or PivotTable.
2. On the **Insert** tab, select Slicer.
3. Insert **Slicer**
4. In the Insert Slicers dialog box, select the **check** boxes for the fields you want to display, then select **OK**.
5. A slicer will be created for every field that you selected. Clicking any of the slicer buttons will automatically apply that filter to the linked table or PivotTable.



Format a slicer

1. You can adjust your slicer preferences in the **Slicer tab** (in newer versions of Excel), or the Design tab (Excel 2016 and older versions) on the ribbon.
2. On the Slicer or Design tab, select a color style that you want.

Make a slicer available in another PivotTable

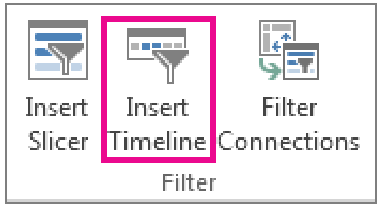
1. First create a PivotTable that is based on the same data source as the PivotTable that already has the slicer want to reuse.
2. Select the slicer that you want to share in another PivotTable. This displays the Slicer tab.
3. On the Slicer tab, select Report Connections.
4. In the dialog box, select the check box of the PivotTable in which you want the slicer to be available.

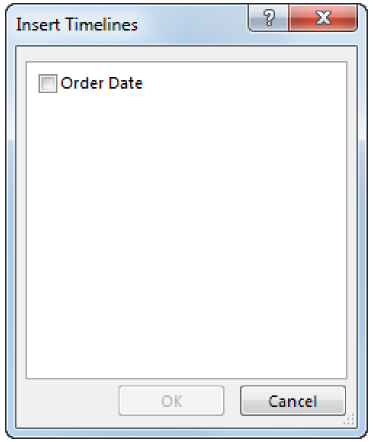
Create a PivotTable timeline to filter dates

Timeline to filter dates

Instead of adjusting filters to show dates, you can use a PivotTable Timeline—a dynamic filter option that lets you easily filter by date/time and zoom in on the period you want with a slider control. Click Analyze > Insert Timeline to add one to your worksheet.

1. Click anywhere in a PivotTable to show the PivotTable Tools ribbon group, then click Analyze > Insert Timeline.
2. In the Insert Timeline dialog box, check the date fields you want, and click OK.





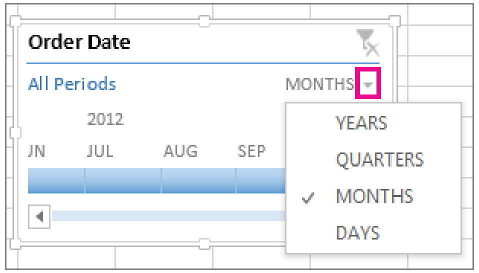
Use a Timeline to filter by time period

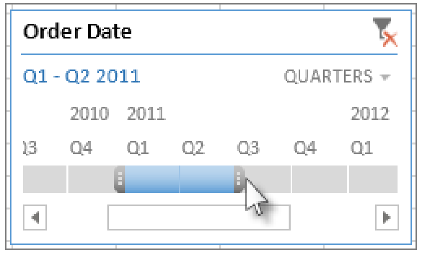
With your Timeline in place, you're ready to filter by a time period in one of four-time levels (years, quarters, months, or days).

1. Click the arrow next to the time level shown and pick the one you want.

2. Drag the Timeline scroll bar to the time period you want to analyze.

3. In the timespan control, click a period tile and drag to include additional tiles to select the date range you want. Use the timespan handles to adjust the date range on either side.

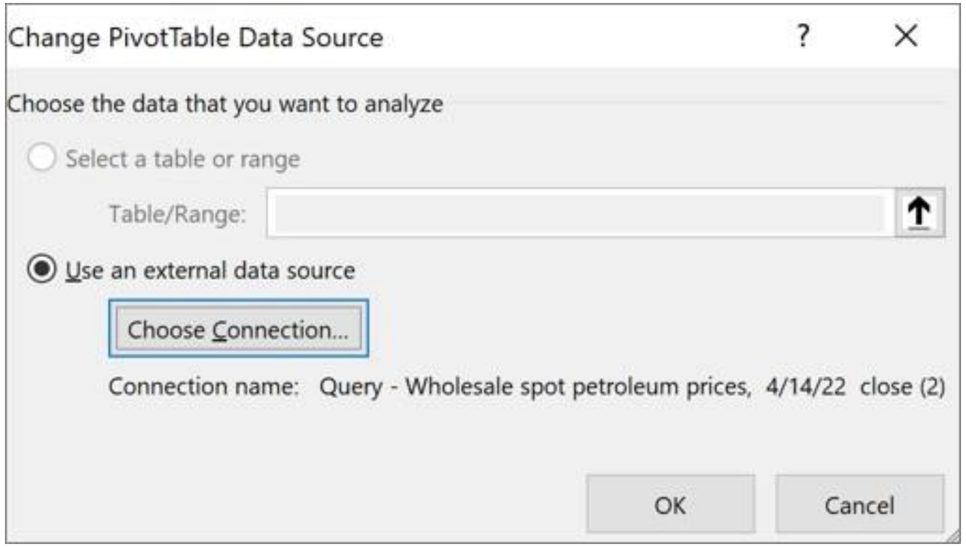




Change the source data for a PivotTable

Change the source data for a PivotTable

- Click anywhere in your PivotTable report to activate it.
- On the **Analyze** tab, in the **Data** group, click **Change Data Source**, and then select **Change Data Source**.
- The **Change PivotTable Data Source** dialog box will appear.
- To use an external data source, click **Use an external data source**, and then click **Choose Connection**.
- The **Existing Connections** dialog box will be displayed.
- In the **Show** drop-down list at the top of the dialog box, select the category of connections you want or choose **All Existing Connections** (the default).
- Select a connection from the **Select a Connection** list box, and then click **Open**.
- **Note**: If your connection is not listed, refer to **Manage Connections to Data in a Workbook** for more information.
- Click **OK** to apply the changes.



1. Note: If you choose a connection from the Connections in this Workbook category, you will be reusing or sharing an existing connection. If you choose a connection from the Connection files on the network or Connection files on this computer category, the connection file is copied into the workbook as a new workbook connection, and then used as the new connection for the PivotTable report.

Q&A

Time to Apply

Hands-on

وزارة الاتصالات وتكنولوجيا المعلومات

رواد مصر الرقمية

Task

- Use an **Excel dataset** that includes fields such as *Region*, *Product Category*, *Sales Amount*, and *Date*.
- Create a **pivot table** to summarize sales by region and product category.
- Use **calculated fields** to compute total and average sales.
- **Group data** by quarter or month to identify time-based trends.

- **Challenge**:
  - Customize the pivot table with **conditional formatting** and **filtering options** to highlight the top-performing product categories in each region.

Thank You